Mass-Spring Oscillation: Finding a Spring Constant Two Ways

SYNOPSIS

You will measure the same spring constant twice, by two independent routes — once by hanging weights on the spring and reading how far it stretches, and once by setting it oscillating and timing it. Neither measurement knows about the other. The point of the lab is what happens when you put the two numbers side by side.

Code LB1-M1 **Due** See the Module 1 Lab folder on Brightspace **Pairs with** Note 4 and Note 5

Where At home, with the lab kit **Report** Full lab report — use the Lab Template

What you need

- Tension spring
- Steel rod, or a hanger that will bridge two supports
- Slotted mass set with hanger
- Ruler, reading to a hundredth of a centimetre
- Stopwatch
- A scale that reads to 1 g, for the spring's own mass

In a teaching lab this is done on a Hooke's Law apparatus. The kit version is the same experiment with simpler parts: the rod rests between two tall objects — chairs, tables, a stack of books — and the spring hangs from it. What matters is that the spring and the masses **hang freely** and that you can get a ruler to the spring.

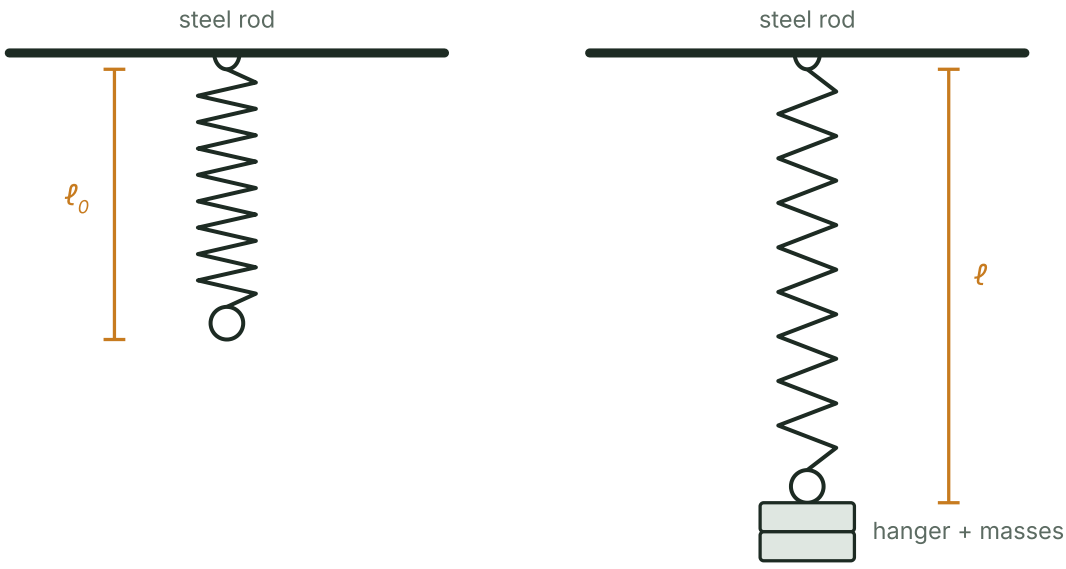


Figure 1. Unloaded and loaded. **Both lengths are measured between the same two points** — top of the hook to the bottom of the loop. If you measure ℓ_0 one way and ℓ another, the difference between them is partly your own inconsistency, and that error goes straight into k.

Theory

A linear spring pushes back in proportion to how far it is deformed:

$$F = kx \quad (1)$$

F is the spring force in N, k the spring constant in N/m, x the deformation in m. The minus sign that normally sits in front of k — saying force and deformation point opposite ways — is dropped here, because every quantity in this lab is a magnitude.

The same system has a second equation, relating how fast it oscillates to the spring constant and the mass:

$$f = 1/T = (1/2\pi)\sqrt{(\, k \,/\, (m + m')\,)} \quad (2)$$

f is the frequency in Hz, T the period in s, m the suspended mass in kg — hanger plus added masses — and m' the **effective spring mass**, which is the share of the spring's own mass that moves with the load.

Solve (2) for T and square both sides:

$$T^2 = (4\pi^2/k)m + (4\pi^2/k)m' \quad (3)$$

Read equation (3) as the equation of a straight line before you take any data

It is $y = Ax + B$ with $y = T^2$ and $x = m$. So a plot of T^2 against m should be a straight line whose **slope is $4\pi^2/k$** and whose **intercept is $4\pi^2 m'/k$**. That is the whole analysis of the dynamic half, decided before the stopwatch comes out: you are not going to average ten values of k, you are going to fit a line and read k off its slope. Knowing which quantity you are plotting against which, and what the slope will mean, is the part of a lab that is done at the desk.

Part I — the static route

1. Set up the rod and spring as in Figure 1.
2. Measure the **undeformed** spring length to a hundredth of a centimetre, from the top of the hook to the bottom of the loop. Call it ℓ_0.
3. Make a raw data table for suspended mass m [g] and deformed length ℓ [cm], like Figure 3. Label the table and both columns, units included.
4. Attach the hanger and add masses in 10 g steps until the spring *starts to deform*. That total — hanger plus added mass — is your first m.
5. Measure the deformed length ℓ between the same two points as in step 2.
6. Repeat in 10 g increments until you have **ten trials**.
7. Convert the masses to kg and compute the weight $W = mg$ in N, with $g = 9.81$ m/s². Put this in a processed table like Figure 4.
8. Compute the elongation $x = \ell - \ell_0$ in m, into the same table.
9. Plot W [N] on the vertical against x [m] on the horizontal, as a scatter plot.
10. Add a linear trendline and display its equation. By equation (1), **the slope is the spring constant**. Record it as k_s, in N/m.

Trial	m [g]	ℓ [cm]
1		
2		
⋮		

10		

Figure 3. Raw data, Part I.

Trial	W [N]	x [m]
1		
2		
⋮		
10		

Figure 4. Processed data, Part I.

Part II — the dynamic route

11. Measure the **spring's own mass** to within 1 g and record it as m_s. You will need it at the very end, and only then.

12. Keep the same setup. Load the hanger to about 140 g total. Adjust if your spring wants it — but **never load it so far that it does not return to ℓ_0**. A permanently stretched spring ends the lab.

13. Pull down a *small* distance and release carefully, straight down, with no sideways swing. Time **twenty complete oscillations** and call it t_1. Do it twice more for t_2 and t_3.

> One complete oscillation is pulled-down → top → back to pulled-down. Counting half-cycles is the commonest way to come out of this lab with a k that is wrong by a factor of four, because T is squared.

14. Record m [g] with t_1, t_2, t_3 [s] in a table like Figure 5.

15. Repeat in 10 g increments for **ten trials** in all.

16. Average the three times to get t. Divide by 20 for the period T, then square it. Record m [kg], t [s], T [s] and T^2 [s²] in a table like Figure 6.

17. Plot T^2 [s²] on the vertical against m [kg] on the horizontal.

18. Add a linear trendline with its equation. From the slope and equation (3), get the spring constant and call it k_d.

Trial	m [g]	t_1 [s]	t_2 [s]	t_3 [s]
1				
⋮				

10				

Figure 5. Raw data, Part II.

Trial	m [kg]	t [s]	T [s]	T^2 [s^2]
1				
⋮				
10				

Figure 6. Processed data, Part II.

Putting the two together

19. Compare the two spring constants and report the percent difference:

$$\%\text{Diff} = (k_s - k_d) / k_d \times 100$$

20. From the **intercept** of the Part II graph and equation (3), calculate the effective spring mass. Use the absolute value if the intercept comes out negative. Call it m'_e — experimental.

21. Look up the standard result for effective spring mass as a fraction of the actual spring mass. With that and your m_s from step 11, compute m'_t — theoretical. Report the percent difference between the two, and comment.

What your report must contain

1. **Introduction** — the goal and the procedure that reaches it, 200 ± 20 words.
2. **Data** — the raw tables only (Figures 3 and 5). No processed data here. Label everything.
3. **Analysis** — sample calculations in this order, each with a line saying what it is for: weight W; elongation x; average time t; period T; T^2; the spring constant k_d; and the percent difference. Then the processed tables (4 and 6), both graphs, a labelled sketch or photograph of your setup, and **the derivation of equation (3) from equation (2)**.
4. **Results** — k_s, k_d, the percent difference between them, and the two effective spring masses. Then comment on what you got, naming the likely sources of error.
5. **Conclusion** — restate what you set out to find and what you found.

> **Make the numbers findable.** Every result goes on its own line, in bold, as an equation with its units — not buried

in a sentence. A reader should be able to find k_s and k_d in your report in under five seconds.

Questions for your discussion

I. Which of your two spring constants do you trust more, and on what evidence from your own graphs — not on which method sounds better?

II. The static route never needed the spring's mass and the dynamic route did. Why?

III. If you had miscounted and timed twenty *half*-oscillations, which way would k_d be wrong, and by what factor?

Answer three of these fully rather than all of them briefly. Name the three you chose.

WATCH · THE LAB WALKED THROUGH

Recording to come. Watch it before you set the kit up — the stretching and the timing both have a knack to them.

Files for this lab

HANDOUT · PDF

This lab, printable

take it to the table with the kit.

EXCEL

Data workbook

the four tables, pre-labelled, with both plots set up.

WORD

Report starter

the section headings with an opening sentence for each.

WORD

Grading rubric

read it before you write, not after.

Lab manual rebuilt for FLCC PHY 152, 2026, from the handout issued with this course. The original carries no author attribution. Figure numbering corrected; the percent-difference definition is left as the course has always stated it.